CS-104 Quiz 1

Fall 2026

Name: _____

This is an in-class, closed-book quiz. If you have any questions, please feel free to ask them. When you are finished, you can turn it in and go.

1. (4 points)

Give the value of each Python expression below. If it raises an error instead, name the error (e.g., "IndexError").

```
>>> 'a' + 'b'
```
'ab' — example answer

```
>>> 7 // 2
```

```
>>> 2 ** 5
```

```
>>> [1, 2, 3][5]
```

```
>>> {'a': 1, 'b': 2}['c']
```

2. (6 points)

Write an algorithm (in plain English, not Python code) that prompts the user for a temperature in Fahrenheit and prints the equivalent temperature in Celsius. Use the **Gather-Compute-Use** pattern from class, and note that `celsius = (fahrenheit - 32) * 5 / 9`.

3. (4 points)

Write a code segment that inputs a number of total minutes from the user (as an integer) and prints how many whole hours and remaining minutes that is. Here is an example run:

```
Total minutes: 130
Hours: 2, remaining minutes: 10
```

Note that you should: - implement the prompt as shown (`Total minutes: `). - use `//` and `%` as appropriate to compute the hours and remaining minutes. - print the result in the format shown above.

Just write a code segment; you don't need to include documentation, an algorithm, or test cases.

4. (3 points)

a) Write a statement that creates a list called `colors` containing the strings `'red'`, `'green'`, and `'blue'`.

b) Write a statement that changes the **last** element of `colors` to `'purple'`.

5. (3 points)

Consider the dictionary defined below.

```
prices = {'apple': 1.50, 'banana': 0.75}
```

Write a statement that adds a new entry mapping `'pear'` to `2.00`, or explain why this cannot be done.